Recommendations against Common Cause Failures in packages for AURIX™ TC3xx devices

AURIX™ 32-bit microcontroller family

About this document

Scope and Purpose

The purpose of this document is to provide recommendations on the choice of mission and monitor signal balls and pins when using devices from the AURIX™ TC3xx microcontroller family.

The recommendations are against the Common Cause Failures (CCF) in the packages for the mission and monitor signal balls and pins.

Note:

1. *The CCFs in Die are not in scope of this document.*
2. *CCFs in the supply network are not covered by this document.*
3. *The recommendations in this document complement the recommendations given for all the homogenous redundancies regarding the mission I/O in the Safety Manual **[1]**.*
4. *It is sufficient if the recommendations stated in this document are followed to reduce the risk of in-package failures that can cause the mission and monitor signals to fail during the system operation. However, Infineon can provide additional support for the assignment of the mission and monitor pins in the packages, in case these recommendations cannot be followed.*

Intended audience

This application note is written for system engineers, software engineers, and functional safety managers involved in the design or development of a safety-related system, who are considering integrating the AURIX™ TC3xx microcontroller hardware as a safety element out of context (SEooC) into their system.

Conventions used

Abbreviation	Acronyms
CCF	Common Cause Failure
I/O	Input/output
LFBGA	Low Fine pitch Ball Grid Array
LQFP	Low Profile Quad Flat Package
PCB	Printed Circuit Board
SEooC	Safety Element out of Context
TQFP	Thin Quad Flat Package

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1 Introduction

The Common Cause Failures (CCF) in the package can affect both the mission and monitor input/output (I/O) signals, potentially resulting in failures. To reduce the risk of failures, recommendations on the choice of pins for the mission signals and monitoring signals are given in this document.

1.1 Common Cause Failures

CCFs are the failure of two or more elements of an item resulting from a single specific event or root cause [2].

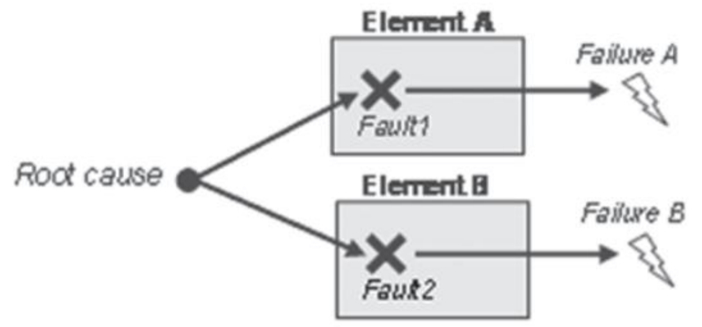


Figure 1. Common Cause Failure (see [2])

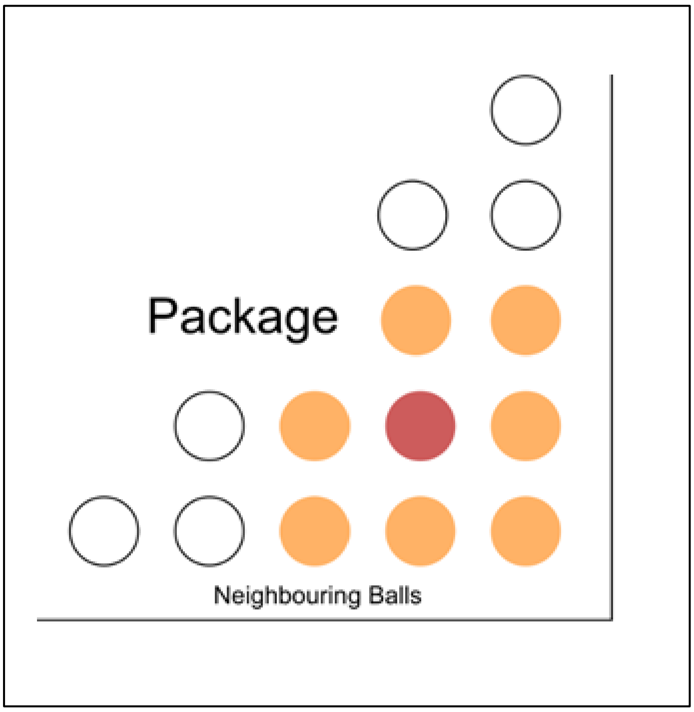
2 Common Cause Failures in LFBGA package

The Common Cause Failure can occur at one of the following parts of the package:

1. Neighboring balls

   This illustrative example shows how a common cause failure can affect the neighboring balls at the package level.

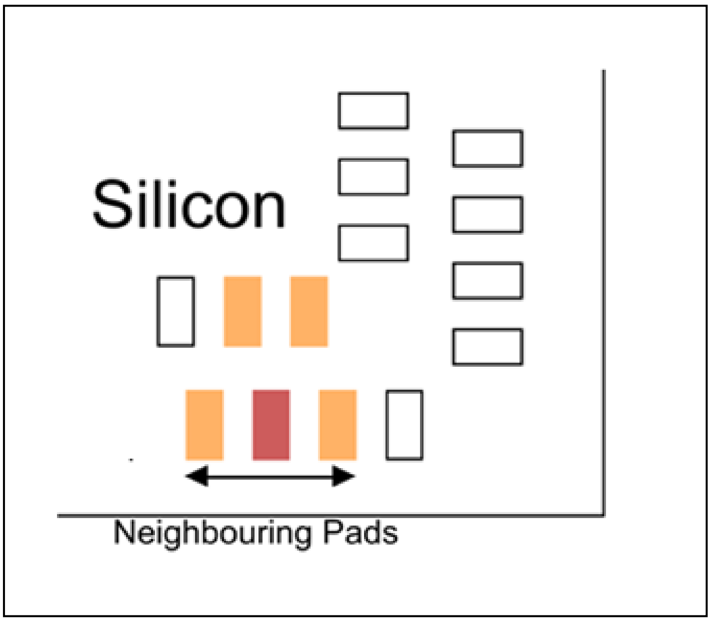
   The balls in consideration is marked in red color while the affected balls are marked in orange color.



2. Neighboring pads

   This illustrative example how a common cause failure can affect the neighboring pads on the Silicon.

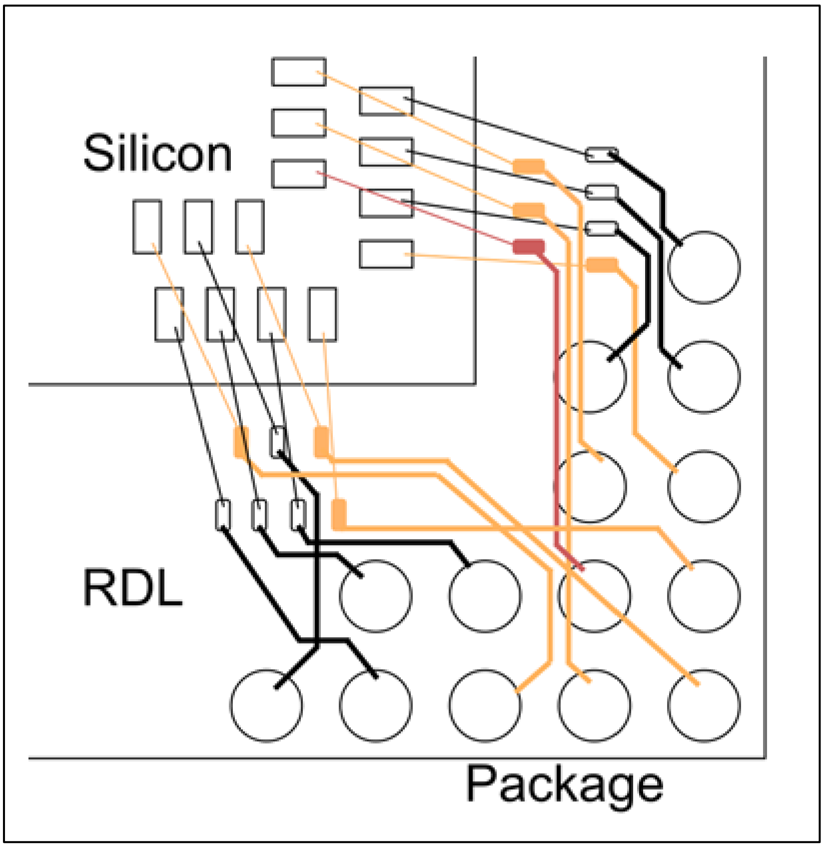
   The pad in consideration is marked in red color while the affected pads are marked in orange color.



3. In the package redistribution layer

This illustrative example shows how a common cause failure can propagate through the re-distribution layer (RDL) and can affect other components as well.

The substrate routing in the re-distribution layer in consideration is marked in red color while the affected RDL routings are marked in orange color.



Note: It is important to understand that the neighboring balls are not always routed to neighboring pads. Therefore the proximity between the mission and monitor input/output signal at the balls, redistribution layer and pads, has been considered for the recommendations given.

2.1 I/O ball configuration in a LFBGA516 package

The ball-out of the package LFBGA516 is separated into 8 different groups. The mission I/O pin can be used from one group and the monitor signal can be used from any other group as indicated in the truth table (see **2.2.1**).

2.1.1 Truth table

		NET A								
		Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7	Group 8	Core
NET B	Group 1	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE
	Group 2	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE
	Group 3	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE
	Group 4	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
	Group 5	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE
	Group 6	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE
	Group 7	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
	Group 8	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE
	Core	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE

Figure 2. Truth table with the valid I/O pin groups for the mission and monitor signals corresponding to the ball-out groups

- The mission I/O pins and the related nets in the category NET A are sufficiently separated when the monitor I/O pin (net) in the category NET B are chosen from the group marked 'TRUE' in the truth table.
- The 'Core' group is always out of the scope.
- As illustrated in the examples below, if the mission signal is connected to the region highlighted in blue (Group 1), then avoid connecting the monitor signal to the same blue region or from the adjacent group marked in yellow (Group 2) and the group marked in purple (Group 8).

2.1.2 TC399 I/O ball configuration in the LFBGA516 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

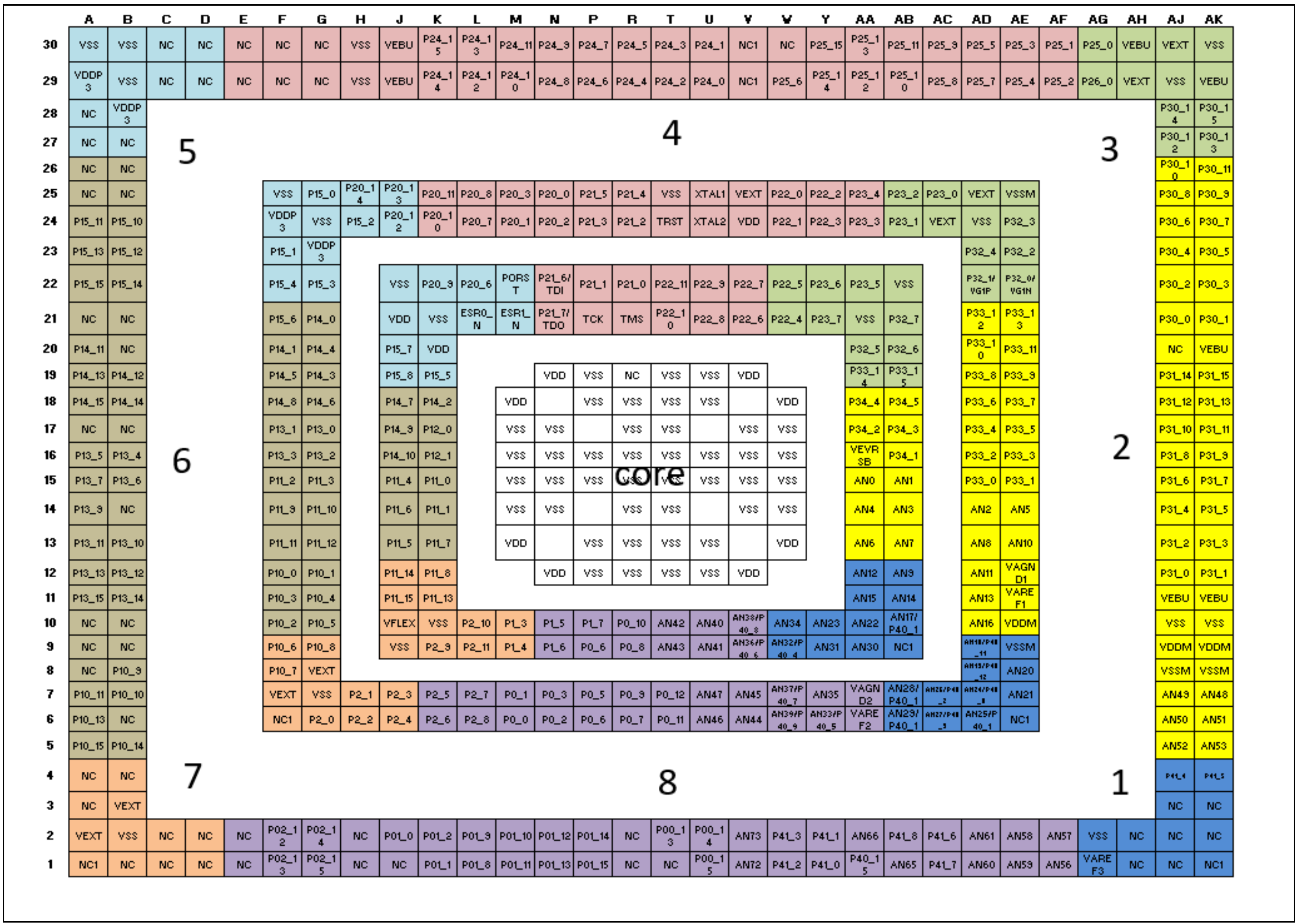


Figure 3. TC399 I/O configuration

See references: [1][2][3][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

Note: The exceptions for using the mission and the monitor signals have been documented in the appendix; ***Exceptions for TC399, BGA516 package****.*

2.1.3 TC389 I/O ball configuration in the LFBGA516 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

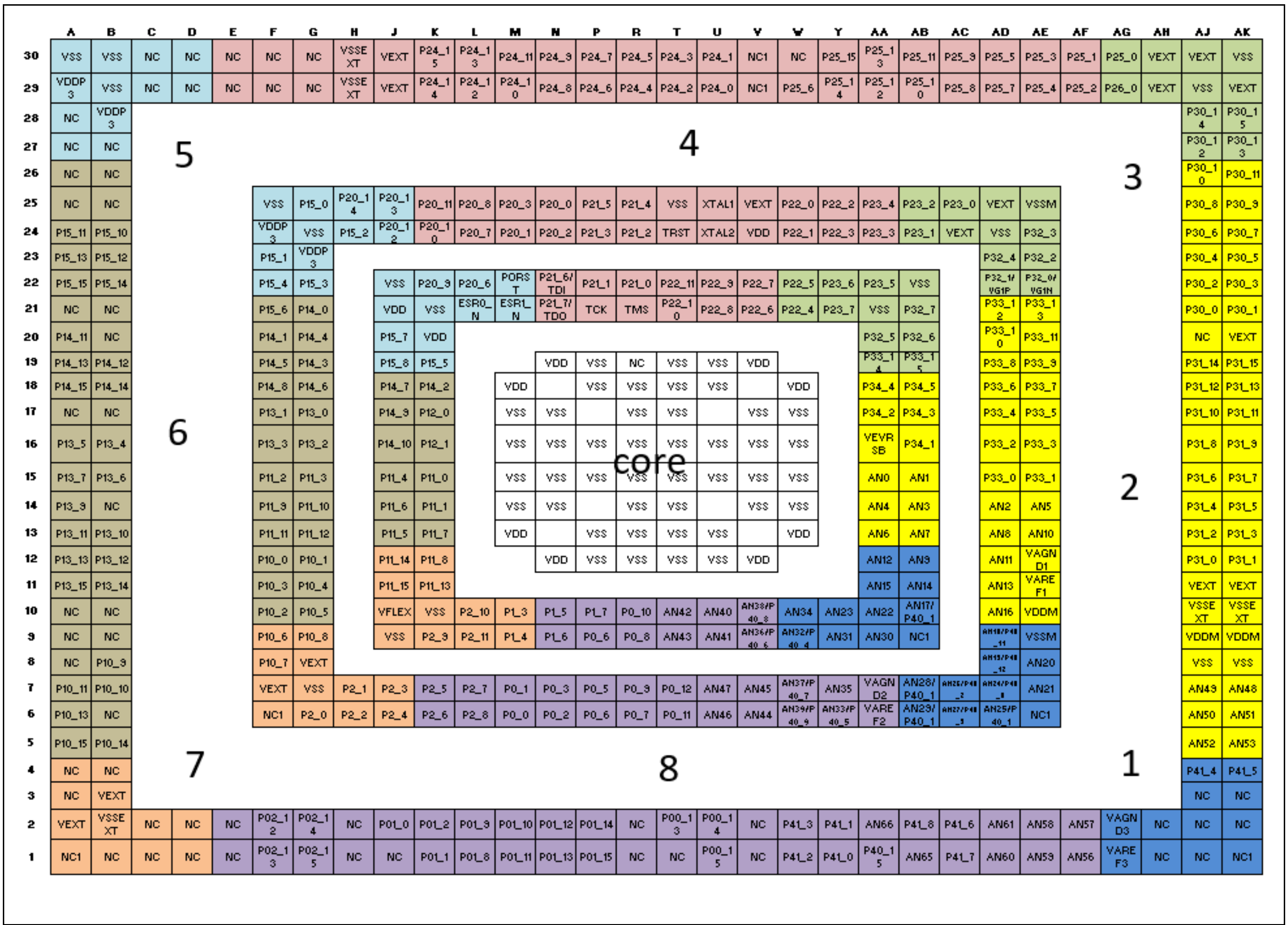


Figure 4. TC389 I/O configuration

See references: [1][2][4][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the mission signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

Note: The exceptions for using the mission and the mission signals have been documented in the appendix; ***Exceptions for TC389, BGA516 package.***

2.2 I/O ball configuration in a LFBGA292 package

The ball-out of the package BGA292 is separated into 8 different groups. The mission I/O pin can be used from one group and the mission signal can be used from any other group as indicated in the truth table (see 2.2.1).

2.2.1 Truth table

		NET A								
		Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7	Group 8	Core
NET B	Group 1	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE
	Group 2	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE
	Group 3	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE
	Group 4	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
	Group 5	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE
	Group 6	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE
	Group 7	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
	Group 8	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE
	Core	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE

Figure 5. Truth table with the valid I/O pin groups for the mission and monitor signals corresponding to the ball-out groups

- The mission I/O pins and the related nets in the category NET A are sufficiently separated when the monitor I/O pin (net) in the category NET B are chosen from the group marked 'TRUE' in the truth table.
- The 'Core' group is always out of the scope.
- As illustrated in the examples below, if the mission signal is connected to the region highlighted in blue (Group 1), then avoid connecting the monitor signal to the same blue region or from the adjacent group marked in yellow (Group 2), and the group marked in purple (Group 8).

2.2.2 TC397 I/O ball configuration in the LFBGA292 package variant

Note: *Regions of different colour indicate the different groups for the mission and monitor signals.*

	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	T	U	V	W	Y
20	VSS	P15_0	P20_14	P20_13	P20_11	P20_8	P20_3	P20_0	P21_5	P21_4	VSS_OSC	XTAL1	VEXT_OSC	P22_0	P22_2	P23_4	P23_2	P23_0	VEXT	VSS
19	VDDP3	VSS	P15_2	P20_12	P20_10	P20_7	P20_1	P20_2	P21_3	P21_2	RST	XTAL2	VDD_OSC	P22_1	P22_3	P23_3	P23_1	VEXT	VSS	P32_3
18	P15_1	VDDP3	Top View																P32_4	P32_2
17	P15_4	P15_3		VSS	P20_9	P20_6	PORST	P21_6/TDI	P21_1	P21_0	P22_11	P22_9	P22_7	P22_5	P23_6	P23_5	VSS		P32_1	P32_0
16	P15_6	P14_0		VDD	VSS	ESR0_N	ESR1_N	P21_7 / TDO	TCK	TMS	P22_10	P22_8	P22_6	P22_4	P23_7	VSS	P32_7		P33_12	P33_13
15	P14_1	P14_4		P15_7	VDD											P32_5	P32_6		P33_10	P33_11
14	P14_5	P14_3		P15_8	P15_5			VDD	VSS	DAPE0	AGBT ERR (VSS)	VSS	VDD			P33_14	P33_15		P33_8	P33_9
13	P14_8	P14_6		P14_7	P14_2		VDD		VSS	VSS	VSS	VSS		VDD		P34_4	P34_5		P33_6	P33_7
12	P13_1	P13_0		P14_9	P12_0		VSS	VSS		VSS	VSS		VSS	VSS		P34_2	P34_3		P33_4	P33_5
11	P13_3	P13_2		P14_10	P12_1		DAPE1	VSS	VSS	VSS	VSS	VSS	VSS	AGBT TXP (Vss)		VEVRSB	P34_1		P33_2	P33_3
10	P11_2	P11_3		P11_4	P11_0		DAPE2	VSS	VSS	VSS	VSS	VSS	VSS	AGBT TXN (Vss)		AN0	AN1		P33_0	P33_1
9	P11_9	P11_10		P11_6	P11_1		VSS	VSS		VSS	VSS		VSS	VSS		AN4	AN3		AN2	AN5
8	P11_11	P11_12		P11_5	P11_7		VDD		VSS	VSS	VSS	VSS		VDD		AN6	AN7		AN8	AN10
7	P10_0	P10_1		P11_14	P11_8			VDDSB	VSS	AGBT CLKP (VSS)	AGBT CLKN (VSS)	VSS	VDD			AN12	AN9		AN11	VAGND1
6	P10_3	P10_4		P11_15	P11_13											AN15	AN14		AN13	VAREF1
5	P10_2	P10_5		VFLEX	VSS_EXT	P2_10	P1_3	P1_5	P1_7	P0_10	AN42	AN40	AN38/P40_8	AN34	AN23	AN22	AN17/P40_10		AN16	VDDM
4	P10_6	P10_8		VSS	P2_9	P2_11	P1_4	P1_6	P0_6	P0_8	AN43	AN41	AN36/P40_6	AN32/P40_4	AN31	AN30	NC1		AN18	VSSM
3	P10_7	VEXT																	AN19/P40_12	AN20
2	VEXT	VSS	P2_1	P2_3	P2_5	P2_7	P0_1	P0_3	P0_5	P0_9	P0_12	AN47	AN45	AN37/P40_7	AN35	VAGND2	AN28/P40_13	AN26/P40_2	[illegible]	AN21
1	NC1	P2_0	P2_2	P2_4	P2_6	P2_8	P0_0	P0_2	P0_4	P0_7	P0_11	AN46	AN44	AN39/P40_9	AN33/P40_5	VAREF2	AN29/P40_14	AN27/P40_3	[illegible]	NC1

Figure 6. TC397 I/O configuration

See references: [1][2][3][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the mission signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

2.2.3 TC397 ADAS I/O ball configuration of the LFBGA292 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

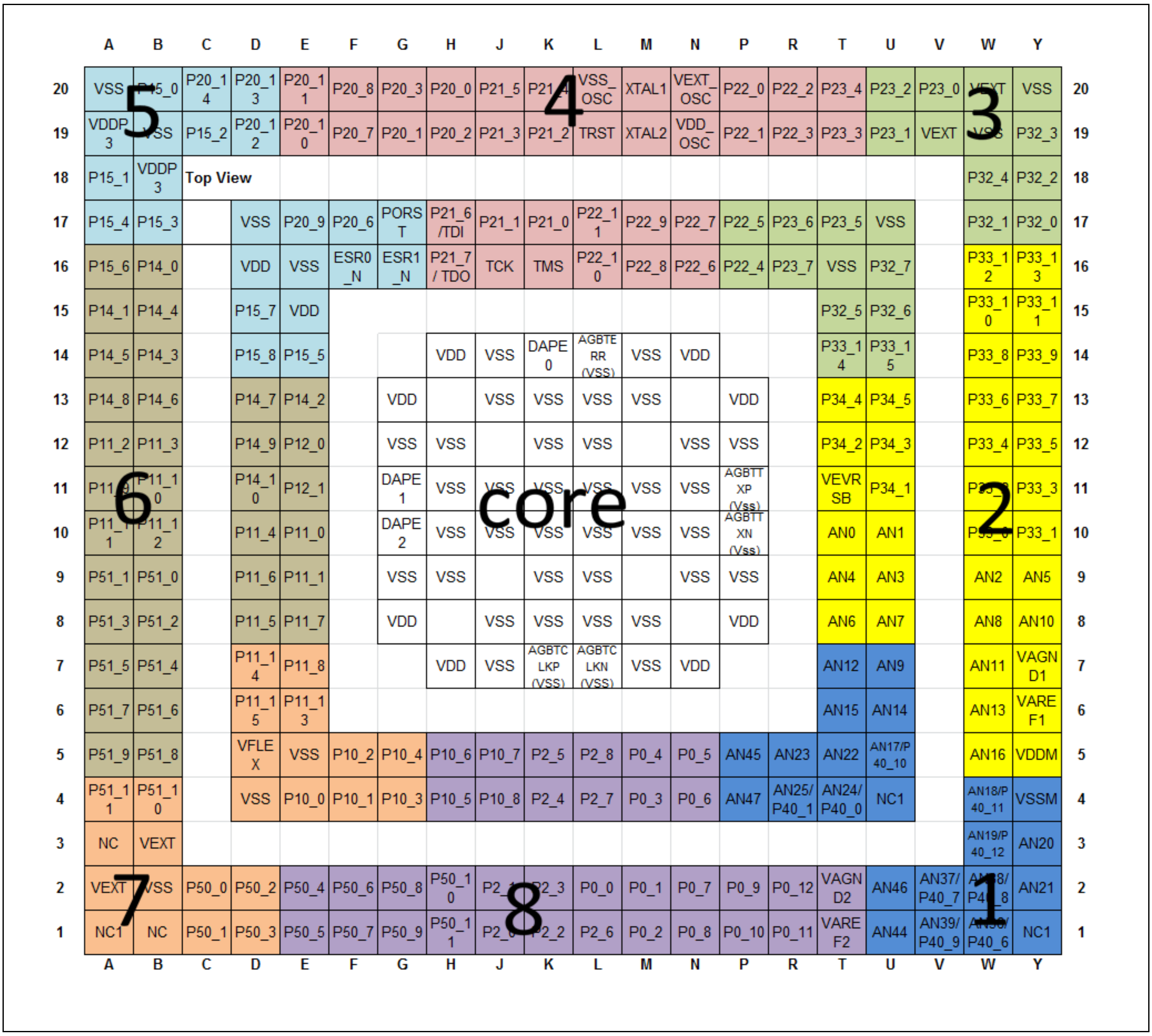


Figure 7. TC397 ADAS I/O configuration

See references: [1][2][3][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

2.2.4 TC387 I/O ball configuration in the LFBGA292 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

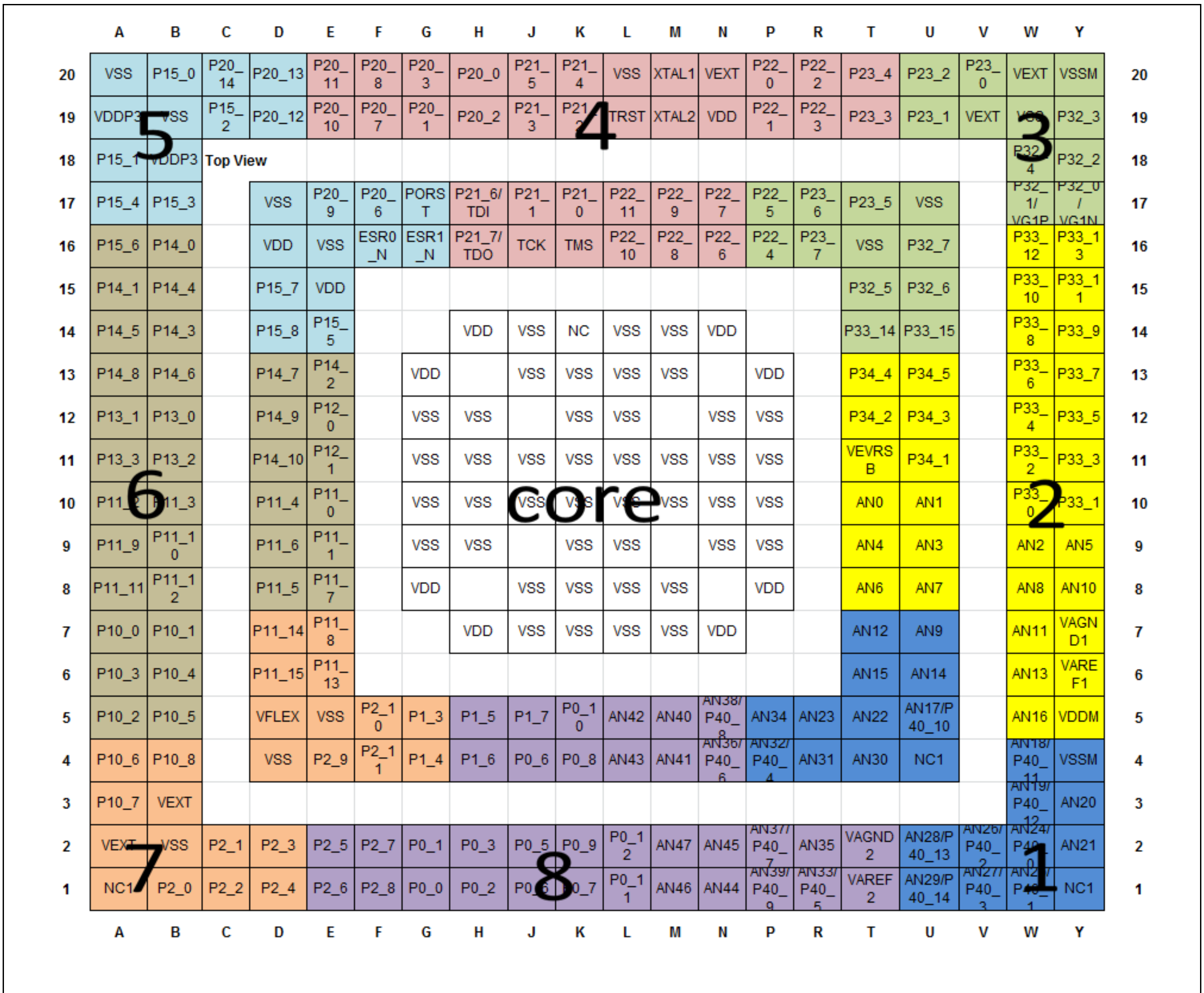


Figure 8. TC387 I/O configurations

See references: [1][2][4][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

2.2.5 TC377 TP I/O ball configuration in the LFBGA292 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	T	U	V	W	Y
20	VSSEXT	P15_0	P20_14	P20_13	P20_11	P20_8	P20_3	P20_0	P21_5	P21_4	VSSOSC	XTAL1	VEXT_OSC	P22_0	P22_2	P23_4	P23_2	P23_0	VEXT	VSSEXT
19	VDDP3	VSSEXT	P15_2	P20_12	P20_10	P20_7	P20_1	P20_2	P21_3	P21_2	RST_N	XTAL2	VDD OSC	P22_1	P22_3	P23_3	P23_1	VEXT	VSSEXT	P32_3
18	P15_1	VDDP3																	P32_4	P32_2
17	P15_4	P15_3		VSSEXT	P20_9	P20_6	PORST_N	P21_6	P21_1	P21_0	P22_11	P22_9	P22_7	P22_5	P23_6	P23_5	VSSEXT		P32_1	P32_0
16	P15_6	P14_0		VDD	VSSEXT	ESR0_N	ESR1_N	P21_7	TCK	TMS	P22_10	P22_8	P22_6	P22_4	P23_7	VSSEXT	P32_7		P33_12	P33_13
15	P14_1	P14_4		P15_7	VDD											P32_5	P32_6		P33_10	P33_11
14	P14_5	P14_3		P15_8	P15_5			VDD	VSS	NC	VSS	VSS	VDD			P33_14	P33_15		P33_8	P33_9
13	P14_8	P14_6		P14_7	P14_2		VDD		VSS	VSS	VSS	VSS		VDD		P34_4	P34_5		P33_6	P33_7
12	P13_1	P13_0		P14_9	P12_0		VSS	VSS		VSS	VSS		VSS	VSS		P34_2	P34_3		P33_4	P33_5
11	P13_3	P13_2		P14_10	P12_1		VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS		VEVRSB	P34_1		P33_2	P33_3
10	P11_2	P11_3		P11_4	P11_0		VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS		AN0	AN1		P33_0	P33_1
9	P11_9	P11_10		P11_6	P11_1		VSS	VSS		VSS	VSS		VSS	VSS		AN4	AN3		AN2	AN5
8	P11_11	P11_12		P11_5	P11_7		VDD		VSS	VSS	VSS	VSS		VDD		AN6	AN7		AN8	AN10
7	P10_0	P10_1		P11_14	P11_8			VDD	VSS	VSS	VSS	VSS	VDD			AN12	AN9		AN11	VSS
6	P10_3	P10_4		P11_15	P11_13											AN15	AN14		AN13	VAREF1
5	P10_2	P10_5		VFLEX	VSSEXT	P02_10	P01_3	P01_5	P01_7	P00_10	AN42	AN40	P40_8	AN34	AN23	AN22	P40_10		AN16	VDDM
4	P10_6	P10_8		VSSEXT	P02_9	P02_11	P01_4	P01_6	P00_6	P00_8	AN43	AN41	P40_6	P40_4	AN31	AN30	NC1		P40_11	VSS
3	P10_7	VEXT																	P40_12	AN20
2	VEXT	VSSEXT	P02_1	P02_3	P02_5	P02_7	P00_1	P00_3	P00_5	P00_9	P00_12	AN47	AN45	P40_7	AN35	VSS	P40_13	P40_2	P40_[illegible]	AN21
1	NC1	P02_0	P02_2	P02_4	P02_6	P02_8	P00_0	P00_2	P00_4	P00_7	P00_11	AN46	AN44	P40_9	P40_5	VAREF2	P40_14	P40_3	P40_[illegible]	NC1

Figure 9. TC377 TP I/O configurations

See references: [1][2][5][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

2.2.6 TC377 TE I/O ball configuration in the LFBGA292 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	T	U	V	W	Y
20	VSSEXT	P15_0	P20_14	P20_13	P20_11	P20_8	P20_3	P20_0	P21_5	P21_4	VSSOSC	XTAL1	VEXT_OSC	P22_0	P22_2	P23_4	P23_2	P23_0	VEXT	VSSEXT
19	VDDP3	VSSEXT	P15_2	P20_12	P20_10	P20_7	P20_1	P20_2	P21_3	P21_2	RST_N	XTAL2	VDDOSC	P22_1	P22_3	P23_3	P23_1	VEXT	VSSEXT	P32_3
18	P15_1	VDDP3																	P32_4	P32_2
17	P15_4	P15_3		VSSEXT	P20_9	P20_6	PORST_N	P21_6	P21_1	P21_0	P22_11	P22_9	P22_7	P22_5	P23_6	P23_5	VSSEXT		P32_1	P32_0
16	P15_6	P14_0		VDD	VSSEXT	ESR0_N	ESR1_N	P21_7	TCK	TMS	P22_10	P22_8	P22_6	P22_4	P23_7	VSSEXT	P32_7		P33_12	P33_13
15	P14_1	P14_4		P15_7	VDD											P32_5	P32_6		P33_10	P33_11
14	P14_5	P14_3		P15_8	P15_5			VDD	VSS	DAPE0	ERR_SIG	VSS	VDD			P33_14	P33_15		P33_8	P33_9
13	P14_8	P14_6		P14_7	P14_2		VDD		VSS	VSS	VSS	VSS		VDD		P34_4	P34_5		P33_6	P33_7
12	P13_1	P13_0		P14_9	P12_0		VSS	VSS		VSS	VSS		VSS	VSS		P34_2	P34_3		P33_4	P33_5
11	P13_3	P13_2		P14_10	P12_1		DAPE1	VSS	VSS	VSS	VSS	VSS	VSS	TX_P		VEVRSB	P34_1		P33_2	P33_3
10	P11_2	P11_3		P11_4	P11_0		DAPE2	VSS	VSS	VSS	VSS	VSS	VSS	TX_N		AN0	AN1		P33_0	P33_1
9	P11_9	P11_10		P11_6	P11_1		VSS	VSS		VSS	VSS		VSS	VSS		AN4	AN3		AN2	AN5
8	P11_11	P11_12		P11_5	P11_7		VDDSB		VSS	VSS	VSS	VSS		VDD		AN6	AN7		AN8	AN10
7	P10_0	P10_1		P11_14	P11_8			VDDSB	VSS	CLK_P	CLK_N	VSS	VDD			AN12	AN9		AN11	VSS
6	P10_3	P10_4		P11_15	P11_13											AN15	AN14		AN13	VAREF1
5	P10_2	P10_5		VFLEX	VSSEXT	P02_10	P01_3	P01_5	P01_7	P00_10	AN42	AN40	P40_8	AN34	AN23	AN22	P40_10		AN16	VDDM
4	P10_6	P10_8		VSSEXT	P02_9	P02_11	P01_4	P01_6	P00_6	P00_8	AN43	AN41	P40_6	P40_4	AN31	AN30	NC1		P40_11	VSS
3	P10_7	VEXT																	P40_12	AN20
2	VEXT	VSSEXT	P02_1	P02_3	P02_5	P02_7	P00_1	P00_3	P00_5	P00_9	P00_12	AN47	AN45	P40_7	AN35	VSS	P40_13	P40_2	P40_0	AN21
1	NC1	P02_0	P02_2	P02_4	P02_6	P02_8	P00_0	P00_2	P00_4	P00_7	P00_11	AN46	AN44	P40_3	P40_5	VAREF2	P40_14	P40_3	P40_1	NC1

Figure 10. TC377 TE I/O configurations

See references: [1][2][6][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

2.2.7 TC377 TX I/O ball configuration in the LFBGA292 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	T	U	V	W	Y
20	VSSEXT	P15_0	P20_14	P20_13	P20_11	P20_8	P20_3	P20_0	P21_5	P21_4	VSSOSC	XTAL1	VEXT_OSC	P22_10	P22_2	P23_4	P23_3	P23_2	VEXT	VSSEXT
19	VDDP3	VSSEXT	P15_2	P20_12	P20_10	P20_7	P20_1	P20_2	P21_3	P21_2	TRST_N	XTAL2	VDDOSC	P22_11	P22_3	P22_12	P23_1	VFLEX2	VSSEXT	P32_3
18	P15_1	VDDP3																	P32_4	P32_2
17	P15_4	P15_3		VSSEXT	P20_9	P20_6	PORST_N	P21_6	P21_1	P21_0	P22_1	P22_9	P22_7	P22_5	P23_6	P23_5	VSSEXT		P32_1	P32_0
16	P15_6	P14_0		VDD	VSSEXT	ESR0_N	ESR1_N	P21_7	TCK	TMS	P22_0	P22_8	P22_6	P22_4	P23_7	VSSEXT	P32_7		P33_12	P33_13
15	P14_1	P14_4		P15_7	VDD											P32_5	P32_6		P33_10	P33_11
14	P14_5	P14_3		P15_8	P15_5			VDD	VSS	DAPE0	ERR_SIG	VSS	VDD			P33_14	P33_15		P33_8	P33_9
13	P14_8	P14_6		P14_7	P14_2		VDD		VSS	VSS	VSS	VSS		VDD		P34_4	P34_5		P33_6	P33_7
12	P13_1	P13_0		P14_9	P12_0		VSS	VSS		VSS	VSS		VSS	VSS		P34_2	P34_3		P33_4	P33_5
11	P13_3	P13_2		P14_10	P12_1		DAPE1	VSS	VSS	VSS	VSS	VSS	VSS	TX_P		VEVRSB	P34_1		P33_2	P33_3
10	P11_2	P11_3		P11_4	P11_0		DAPE2	VSS	VSS	VSS	VSS	VSS	VSS	TX_N		AN0	AN1		P33_0	P33_1
9	P11_9	P11_10		P11_6	P11_1		VSS	VSS		VSS	VSS		VSS	VSS		AN4	AN3		AN2	AN5
8	P11_11	P11_12		P11_5	P11_7		VDDSB		VSS	VSS	VSS	VSS		VDD		AN6	AN7		AN8	AN10
7	P10_0	P10_1		P11_14	P11_8			VDDSB	VSS	CLK_P	CLK_N	VSS	VDD			AN12	AN9		AN11	VSS
6	P10_3	P10_4		P11_15	P11_13											AN15	AN14		AN13	VAREF1
5	P10_2	P10_5		VFLEX	VSSEXT	P02_10	P01_3	P01_5	P01_7	P00_10	AN42	AN40	P40_8	AN34	AN23	AN22	P40_10		AN16	VDDM
4	P10_6	P10_8		VSSEXT	P02_9	P02_11	P01_4	P01_6	P00_6	P00_8	AN43	AN41	P40_6	P40_4	AN31	AN30	NC1		P40_11	VSS
3	P10_7	VEXT																	P40_12	AN20
2	VEXT	VSSEXT	P02_1	P02_3	P02_5	P02_7	P00_1	P00_3	P00_5	P00_9	P00_12	AN47	AN45	P40_7	AN35	VSS	P40_13	P40_2	P40_0	AN21
1	NC1	P02_0	P02_2	P02_4	P02_6	P02_8	P00_0	P00_2	P00_4	P00_7	P00_11	AN46	AN44	P40_9	P40_5	VAREF2	P40_14	P40_3	P40_1	NC1

Figure 11. TC377 TE I/O configurations

See references: [1][2][6][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

2.2.8 TC367 I/O ball configuration in the LFBGA292 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	T	U	V	W	Y
20	VSSEXT	P15_0	P20_14	P20_13	P20_11	P20_8	P20_3	P20_0	P21_5	P21_4	VSSOSC	XTAL1	VEXT_OSC	P22_0	P22_2	P23_4	P23_2	P23_0	VEXT	VSSEXT
19	VDDP3	VSSEXT	P15_2	P20_12	P20_10	P20_7	P20_1	P20_2	P21_3	P21_2	TEST_N	XTAL2	VDDOSC	P22_1	P22_3	P23_3	P23_1	VEXT	VSSEXT	P32_3
18	P15_1	VDDP3																	P32_4	P32_2
17	P15_4	P15_3		VSSEXT	P20_9	P20_6	PORST_N	P21_6	P21_1	P21_0	NC	NC	NC	NC	NC	P23_5	VSSEXT		P32_1	P32_0
16	P15_6	P14_0		VDD	VSSEXT	ESR0_N	ESR1_N	P21_7	TCK	TMS	NC	NC	NC	NC	NC	VSSEXT	NC		P33_12	P33_13
15	P14_1	P14_4		P15_7	VDD											NC	NC		P33_10	P33_11
14	P14_5	P14_3		P15_8	P15_5			VDD	VSS	NC	VSS	VSS	VDD			NC	NC		P33_8	P33_9
13	P14_8	P14_6		P14_7	P14_2		VDD		VSS	VSS	VSS	VSS		VDD		NC	NC		P33_6	P33_7
12	P13_1	P13_0		P14_9	P12_0		VSS	VSS		VSS	VSS		VSS	VSS		NC	NC		P33_4	P33_5
11	P13_3	P13_2		P14_10	P12_1		VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS		VEVRSB	NC		P33_2	P33_3
10	P11_2	P11_3		P11_4	P11_0		VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS		AN0	AN1		P33_0	P33_1
9	P11_9	P11_10		P11_6	P11_1		VSS	VSS		VSS	VSS		VSS	VSS		AN4	AN3		AN2	AN5
8	P11_11	P11_12		P11_5	P11_7		VDD		VSS	VSS	VSS	VSS		VDD		AN6	AN7		AN8	AN10
7	P10_0	P10_1		P11_14	P11_8			VDD	VSS	VSS	VSS	VSS	VDD			AN12	AN9		AN11	VSS
6	P10_3	P10_4		P11_15	P11_13											AN15	AN14		AN13	VAREF1
5	P10_2	P10_5		VFLEX	VSSEXT	NC	NC	NC	NC	P00_10	AN42	AN40	P40_8	AN34	AN23	AN22	AN17		AN16	VDDM
4	P10_6	P10_8		VSSEXT	NC	NC	NC	NC	P00_6	P00_8	AN43	AN41	P40_6	P40_4	AN31	AN30	NC		AN18	VSS
3	P10_7	VEXT																	AN19	AN20
2	VEXT	VSSEXT	P02_1	P02_3	P02_5	P02_7	P00_1	P00_3	P00_5	P00_[illegible]	P00_12	AN47	AN45	P40_7	AN35	NC	AN28	P40_2	P40_0	AN21
1	NC	P02_0	P02_2	P02_4	P02_6	P02_8	P00_0	P00_2	P00_4	P00_7	P00_11	AN46	AN44	P40_9	P40_5	NC	AN29	P40_3	P40_1	NC

Figure 12. TC367 I/O configurations

See references: [1][2][7][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

2.2.9 TC357 I/O ball configuration in the LFBGA292 package variant

Note: *Regions of different colour indicate the different groups for the mission and monitor signals.*

	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	T	U	V	W	Y	
20	VSS	P15_0	P20_14	P20_13	P20_11	P20_8	P20_3	P20_0	P21_5	P21_4	VSS	XTAL1	VEXT	P22_0	P22_2	P23_4	P23_2	P23_0	VEXT	VSSM	20
19	VDDP3	VSS	P15_2	P20_12	P20_10	P20_7	P20_1	P20_2	P21_3	P21_2	TRST	XTAL2	VDD	P22_1	P22_3	P23_3	P23_1	VEXT	VSS	P32_3	19
18	P15_1	VDDP3	Top View																P32_4	P32_2	18
17	P15_4	P15_3		VSS	P20_9	P20_6	PORST	P21_6/TDI	P21_1	P21_0	NC	NC	NC	P22_5	NC	P23_5	VSS		P32_1/VGIP	P32_0/VGIN	17
16	P15_6	P14_0		VDD	VSS	ESR0_N	ESR1_N	P21_7/TDO	TCK	TMS	NC	NC	NC	P22_4	NC	VSS	NC		P33_12	P33_13	16
15	P14_1	P14_4		P15_7	VDD											P32_5	P32_6		P33_10	P33_11	15
14	P14_5	P14_3		P15_8	P15_5			VDD	VSS	DAPE0	AGBT_ERR	VSS	VDD			NC	NC		P33_8	P33_9	14
13	P14_8	P14_6		P14_7	P14_2		VDD		VSS	VSS	VSS	VSS		VDD		NC	NC		P33_6	P33_7	13
12	P11_2	P11_3		P14_9	P12_0		VSS	VSS		VSS	VSS		VSS	VSS		P34_2	P34_3		P33_4	P33_5	12
11	P11_9	P11_10		P14_10	P12_1		DAPE1	VSS	VSS	VSS	VSS	VSS	VSS	AGBT_TX0P		VEVRSB	P34_1		P33_2	P33_3	11
10	P11_11	P11_12		P11_4	P11_0		DAPE2	VSS	VSS	VSS	VSS	VSS	VSS	AGBT_TX0N		AN0	AN1		P33_0	P33_1	10
9	P51_1	P51_0		P11_6	P11_1		VSS	VSS		VSS	VSS		VSS	VSS		AN4	AN3		AN2	AN5	9
8	P51_3	P51_2		P11_5	P11_7		VDD		VSS	VSS	VSS	VSS		VDD		AN6	AN7		AN8	AN10	8
7	P51_5	P51_4		P11_14	P11_8			VDD	VSS	AGBT_CLKP	AGBT_CLKN	VSS	VDD			AN12	AN9		AN11	VAGND1	7
6	P51_7	P51_6		P11_15	P11_13											AN15	AN14		AN13	VAREF1	6
5	P51_9	P51_8		VFLEX	VSS	P10_2	P10_4	P10_6	P10_7	P2_5	P2_8	P0_4	P0_5	NC	NC	NC	NC		NC	VDDM	5
4	P51_11	P51_10		VSS	P10_0	P10_1	P10_3	P10_5	P10_8	P2_4	P2_7	P0_3	P0_6	NC	NC	NC	NC1		NC	VSSM	4
3	NC	VEXT																	NC	NC	3
2	VEXT	VSS	P50_0	P50_2	P50_4	P50_6	P50_8	P50_10	P2_1	P2_3	P0_0	P0_1	P0_7	P0_9	P0_12	NC	NC	NC	NC	NC	2
1	NC1	NC	P50_1	P50_3	P50_5	P50_7	P50_9	P50_11	P2_0	P2_2	P2_6	P0_2	P0_8	P0_10	P0_11	NC	NC	NC	NC	NC1	1
	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	T	U	V	W	Y	

Figure 13. TC357 I/O configuration

See references: [1][2][8][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

Note: *The additional information for choosing the mission and the monitor signals has been documented in appendix;* ***Additional hint for TC357, BGA292 package***

2.2.10 TC337 TP I/O ball configuration in the LFBGA292 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	T	U	V	W	Y
20	VSSEXT	P15_0	P20_14	P20_13	P20_11	P20_8	P20_3	P20_0	P21_5	P21_4	VSSOSC	XTAL1	VEXT_OSC	P22_0	P22_2	NC	NC	NC	VEXT	VSSEXT
19	VDDP3	VSSEXT	P15_2	P20_12	P20_10	P20_7	P20_1	P20_2	P21_3	P21_2	TEST_N	XTAL2	VDDOSC	P22_1	P22_3	NC	P23_1	VEXT	VSSEXT	NC
18	P15_1	VDDP3																	P32_4	NC
17	P15_4	P15_3		VSSEXT	P20_9	P20_6	PORST_N	P21_6	P21_1	P21_0	NC	NC	NC	NC	NC	P23_5	VSSEXT		VCAPP	VCAPN
16	P15_6	P14_0		VDD	VSSEXT	ESR0_N	ESR1_N	P21_7	TCK	TMS	NC	NC	NC	P22_4	NC	VSSEXT	NC		P33_12	NC
15	P14_1	P14_4		P15_7	VDD											NC	NC		P33_10	P33_11
14	P14_5	P14_3		P15_8	P15_5			VDD	VSS	NC	VSS	VSS	VDDO_BGA			NC	NC		P33_8	P33_9
13	P14_8	P14_6		P14_7	P14_2		VDD		VSS	VSS	VSS	VSS		VDDO_BGA		NC	NC		P33_6	P33_7
12	P13_1	P13_0		P14_9	NC		VSS	VSS		VSS	VSS		VSS	VSS		P34_2	P34_3		P33_4	P33_5
11	P13_3	P13_2		P14_10	NC		VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS		VEVRSB	P34_1		P33_2	P33_3
10	P11_2	P11_3		NC	NC		VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS		AN0	AN1		P33_0	P33_1
9	P11_9	P11_10		P11_6	NC		VSS	VSS		VSS	VSS		VSS	VSS		AN4	AN3		AN2	AN5
8	P11_11	P11_12		NC	NC		VDD		VSS	VSS	VSS	VSS		VDD		AN6	AN7		AN8	AN10
7	P10_0	P10_1		NC	P11_8			VDD	VSS	VSS	VSS	VSS	VDD			AN12	AN9		AN11	VSS
6	P10_3	P10_4		NC	NC											AN15	AN14		AN13	VAREF1
5	P10_2	P10_5		VFLEX	VSSEXT	NC	NC	NC	NC	P00_10	NC	NC	P40_8	AN34	NC	NC	NC		NC	VDDM
4	P10_6	P10_8		VSSEXT	NC	NC	NC	NC	P00_6	P00_8	NC	NC	P40_6	P40_4	NC	NC	NC		NC	VSS
3	P10_7	VEXT																	NC	NC
2	VEXT	VSSEXT	P02_1	P02_3	P02_5	P02_7	P00_1	P00_3	P00_5	P00_[illegible]	P00_12	NC	NC	P40_7	AN35	NC	NC	NC	NC	NC
1	NC	P02_0	P02_2	P02_4	P02_6	P02_8	P00_0	P00_2	P00_4	P00_7	P00_11	NC	NC	P40_9	P40_5	NC	NC	NC	NC	NC

Figure 14. TC337 TP I/O configurations

See references: [1][2][9][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.
- Exceptional case:
  - Ball R1 from group 8 and ball W6 from group 2 is the only pair pair (indicated by the box with a red border) which are not sufficiently separated. Therefore it is recommended not to use them together for the monitor signal and mission signal.

2.2.11 TC337 EXT I/O ball configuration in the LFBGA292 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	T	U	V	W	Y
20	VSSEXT	P15_0	P20_14	P20_13	P20_11	P20_8	P20_3	P20_0	P21_5	P21_4	VSSOSC	XTAL1	VEXT_OSC	P22_0	P22_2	NC	NC	NC	VEXT	VSSEXT
19	VDDP3	VSSEXT	P15_2	P20_12	P20_10	P20_7	P20_1	P20_2	P21_3	P21_2	TRST_N	XTAL2	VDDOSC	P22_1	P22_3	NC_T19	P23_1	VEXT	VSSEXT	NC
18	P15_1	VDDP3																	P32_4	NC
17	P15_4	P15_3		VSSEXT	P20_9	P20_6	PORST_N	P21_6	P21_1	P21_0	NC	NC	NC	P22_5	NC	P23_5	VSSEXT		P32_1	P32_0
16	P15_6	P14_0		VDD	VSSEXT	ESR0_N	ESR1_N	P21_7	TCK	TMS	NC	NC	NC	P22_4	NC	VSSEXT	NC		P33_12	P33_13
15	P14_1	P14_4		P15_7	VDD											NC	NC		P33_10	P33_11
14	P14_5	P14_3		P15_8	P15_5			VDD	VSS	DAPE0	ERR_SIG	VSS	VDD			NC	NC		P33_8	P33_9
13	P14_8	P14_6		P14_7	P14_2		VDD		VSS	VSS	VSS	VSS		VDD		NC	NC		P33_6	P33_7
12	P11_2	P11_3		P14_9	P12_0		VSS	VSS		VSS	VSS		VSS	VSS		P34_2	P34_3		P33_4	P33_5
11	P11_9	P11_10		P14_10	P12_1		DAPE1	VSS	VSS	VSS	VSS	VSS	VSS	TX_P		VEVRSB	P34_1		P33_2	P33_3
10	P11_11	P11_12		P11_4	P11_0		DAPE2	VSS	VSS	VSS	VSS	VSS	VSS	TX_N		AN0	AN1		P33_0	P33_1
9	NC	NC		P11_6	P11_1		VSS	VSS		VSS	VSS		VSS	VSS		AN4	AN3		AN2	AN5
8	NC	NC		P11_5	P11_7		VDD		VSS	VSS	VSS	VSS		VDD		AN6	AN7		AN8	AN10
7	NC	NC		P11_14	P11_8			VDD	VSS	CLK_P	CLK_N	VSS	VDD			AN12	AN9		AN11	VSS
6	NC	NC		P11_15	P11_13											AN15	AN14		AN13	VAREF1
5	NC	NC		VFLEX	VSSEXT	P10_2	P10_4	P10_6	P10_7	P02_5	P02_8	P00_4	P00_5	NC	AN23	AN22	AN17		AN16	VDDM
4	NC	NC		VSSEXT	P10_0	P10_1	P10_3	P10_5	P10_8	P02_4	P02_7	P00_3	P00_6	NC	NC	NC	NC		AN18	VSS
3	NC	VEXT																	AN19	AN20
2	VEXT	VSSEXT	P50_0	P50_2	P50_4	P50_6	P50_8	P50_10	P02_1	P02_3	P00_0	P00_1	P00_7	P00_9	P00_12	NC	NC	NC	NC	AN21
1	NC1	NC	P50_1	P50_3	P50_5	P50_7	P50_9	P50_11	P02_0	P02_2	P02_6	P00_2	P00_8	NC	NC	NC	NC	NC	NC	NC

Figure 15. TC337 EXT I/O configurations

See references: [1][2][10][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

Note: The additional information for choosing the mission and the monitor signals has been documented in appendix; ***Additional hint for TC337 EXT,*** *BGA292 package*

2.2.12 TC3E7 I/O ball configuration in the LFBGA292 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

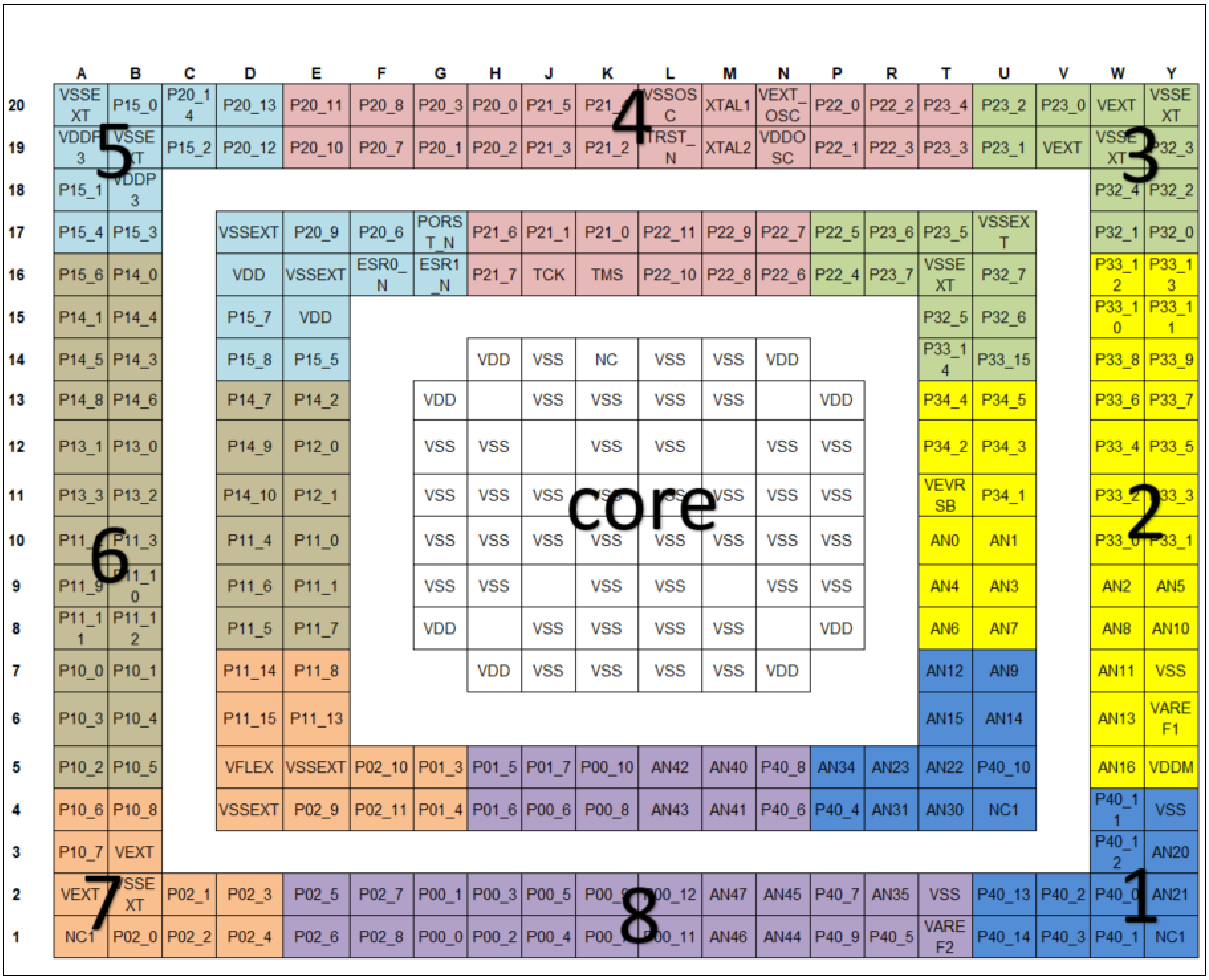


Figure 16. TC3E7 I/O configurations

See references: [1][2][11][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

2.3 I/O ball configuration in a LFBGA180 package

The BGA180 is almost a fully populated BGA package. To provide ease of use, BGA180 devices have been handled diffrerently to the previous packages.

The mission I/O pin can be used from one group and the monitor signal can be used from any other group as indicated in the truth table (see 2.3.1)

2.3.1 Truth table

NET B \ NET A	Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7	Group 8	Group 9	Group 10	Group 11	Group 12	Core
Group 1	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE
Group 2	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE
Group 3	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE
Group 4	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE
Group 5	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE
Group 6	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE
Group 7	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE
Group 8	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
Group 9	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE
Group 10	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE
Group 11	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
Group 12	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE
Core	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE

Figure 17. Truth table with the valid I/O pin groups for the mission and monitor signals corresponding to the ball-out groups

- The mission I/O pins and the related nets in the category NET A are sufficiently separated when the monitor I/O pin (net) in the category NET B are chosen from the group marked 'TRUE' in the truth table.
- The 'Core' group is always out of the scope.
- As illustrated in the examples below, if the mission signal is connected to the region highlighted in light blue (Group 11), then avoid connecting the monitor signal to the same blue region or from the adjacent dark green (Group 10) and the group marked in light green (Group 12).
- In the examples illustrated below, the gray zone consists of the NC, Voltage balls and system I/O balls, and is kept out of scope.

2.3.2 TC366 I/O ball configuration in the LFBGA180 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

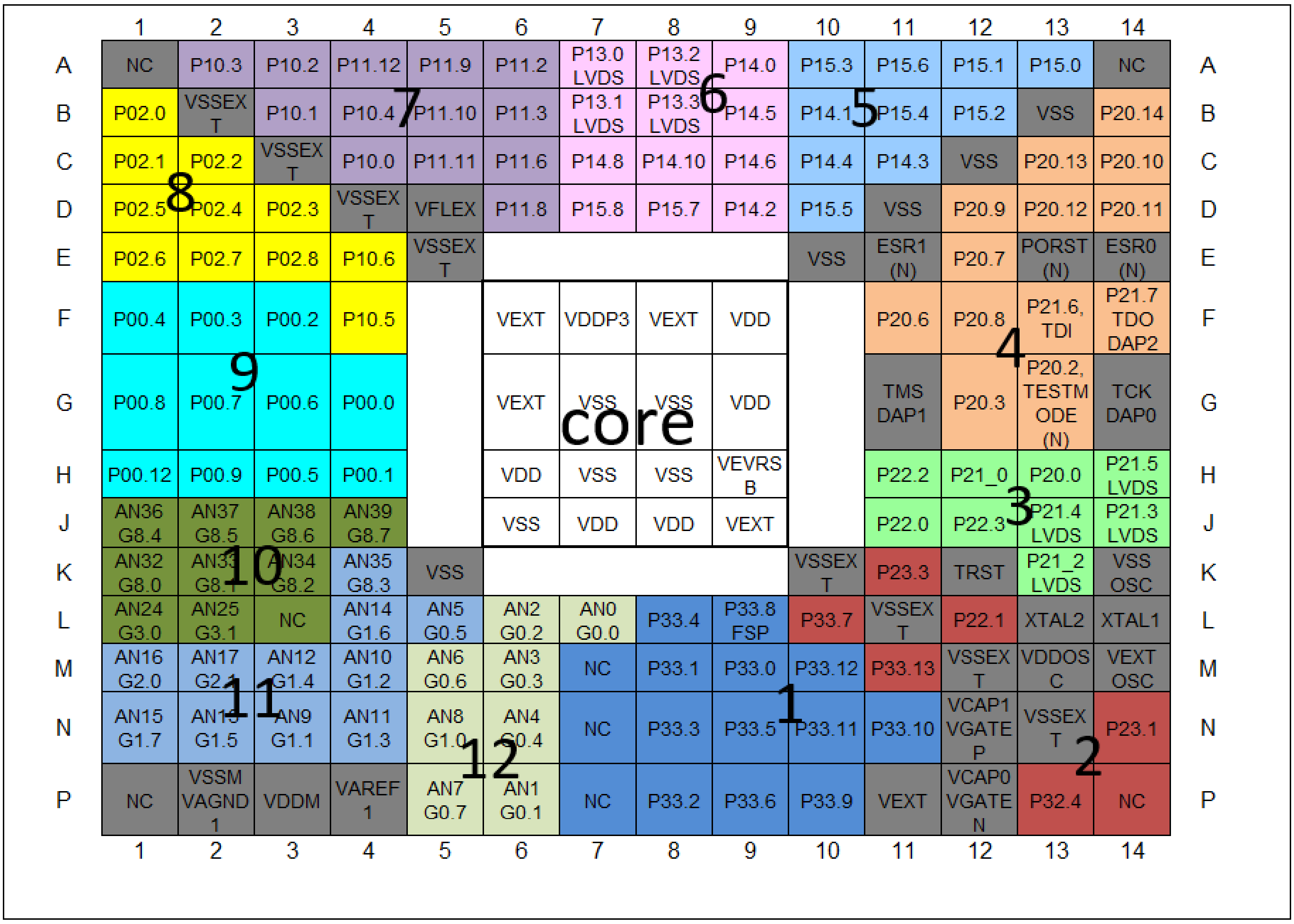


Figure 18. TC366 I/O configurations

See references: [1][2][7][12]

- Twelve different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

2.3.3 TC336 I/O ball configuration in the LFBGA180 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
A	NC	P10.3	P10.2	P11.12	P11.9	P11.2	P13.0 LVDS	P13.2 LVDS	P14.0	P15.3	P15.6	P15.1	P15.0	NC	A
B	P02.0	VSSEXT	P10.1	P10.4	P11.10	P11.3	P13.1 LVDS	P13.3 LVDS	P14.5	P14.1	P15.4	P15.2	VSS	P20.14	B
C	P02.1	P02.2	VSSEXT	NC	P11.11	P11.6	NC	NC	P14.6	P14.4	P14.3	VSS	P20.13	P20.10	C
D	P02.5	P02.4	P02.3	VSSEXT	VEXT	P11.8	P15.8	P15.7	P14.2	P15.5	VSS	P20.9	P20.12	P20.11	D
E	P02.6	P02.7	P02.8	P10.6	VSSEXT					VSS	ESR1 (N)	P20.7	PORST (N)	ESR0 (N)	E
F	P00.4	P00.3	P00.2	P10.5		VEXT	VDDP3	VEXT	VDD		P20.6	P20.8	P21.6, TDI	P21.7 TDO DAP2	F
G	P00.8	P00.7	P00.6	P00.0		VEXT	VSS	VSS	VDD		TMS DAP1	P20.3	P20.2, TESTMODE (N)	TCK DAP0	G
H	P00.12	P00.9	P00.5	P00.1		VDD	VSS	VSS	VEVRSB		P22.2	P21_0	P20.0	P21.5 LVDS	H
J	AN36 G8.4	AN37 G8.5	AN38 G8.6	AN39 G8.7		VSS	VDD	VDD OUT	VEXT		P22.0	P22.3	P21.4 LVDS	P21.3 LVDS	J
K	AN32 G8.0	AN33 G8.1	AN34 G8.2	AN35 G8.3	VSS					VSSEXT	NC	TRST (N)	P21_2 LVDS	VSS OSC	K
L	NC	NC	NC	AN14 G1.6	AN5 G0.5	AN2 G0.2	AN0 G0.0	P33.4	P33.8 FSP	P33.7	VSSEXT	P22.1	XTAL2	XTAL1	L
M	NC	NC	AN12 G1.4	AN10 G1.2	AN6 G0.6	AN3 G0.3	P34.1	P33.1	P33.0	P33.12	NC	VSSEXT	VDDOSC	VEXT OSC	M
N	AN15 G1.7	AN13 G1.5	AN9 G1.1	AN11 G1.3	AN8 G1.0	AN4 G0.4	P34.2	P33.3	P33.5	P33.11	P33.10	VCAP1 VGATEP	VSSEXT	P23.1	N
P	NC	VSSM VAGND1	VDDM	VAREF1	AN7 G0.7	AN1 G0.1	P34.3	P33.2	P33.6	P33.9	VEXT	VCAP0 VGATEN	NC	NC	P
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	

Figure 19. TC336 I/O configurations

See references: [1][2][9][12]

- Twelve different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal. However, it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.

2.4 I/O Ball configuration in a LFBGA180 ADAS package

The BGA180 is almost a fully populated BGA package. To provide ease of use, BGA180 devices are handled diffrerently to the previous packages.

The mission I/O pin can be used from one group and the monitor signal can be used from any other group as indicated in the truth table (see **2.4.1**)

2.4.1 Truth table

		Mission Signal									
		Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7	Group 8	Group 9	Core
Monitor Signal	Group 1	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE
	Group 2	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE
	Group 3	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE
	Group 4	TRUE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE
	Group 5	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
	Group 6	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE
	Group 7	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE
	Group 8	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
	Group 9	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE
	Core	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE

Figure 20. Truth table with the valid I/O pin groups for the mission and monitor signals corresponding to the ball-out groups

- The mission I/O pins and the related nets in the category NET A are sufficiently separated when the monitor I/O pin (net) in the category NET B are chosen from the group marked 'TRUE' in the truth table.
- The 'Core' group is always out of the scope.
- Due to large availibility of NC balls and the voltage balls, there is sufficient spacing between the adjacent groups. For example; Group 1–Group 2, or Group 6–Group 7. Therefore it is recommended to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.
  - But this is not true for other group pairs! For example; Group 3-Group 4 and Group 5-Group 6. Therefore it is recommended not to use the balls in the adjacent regions for the monitor signals, when using the balls from one group for the mission signal.
- As illustrated in the examples that follow, if the mission signal is connected to the region highlighted in green (Group 3), then avoid connecting the monitor signal to the same green region or from the adjacent red (Group 4) and yellow green (Group 2) groups.
- In the examples illustrated below, the gray zone consists of the NC, Voltage balls and system I/O balls, and is kept out of scope.

2.4.2 TC356 ADAS I/O ball configuration in the LFBGA180 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
A	NC	NC	P11.10 RXD0	P11.9 RXD1	P11.6 TXEN	P11.2 TXD1	P11.0 TXD3	P12.1	P14.0	P15.3	P15.6	P15.1	P15.0	NC	A
B	NC	VSSEXT	P11.8 RXD2	P11.7 RXD3	P11.5 TXCLK	P11.4 RXCLK	P11.1 TXD2	P11.14	P14.5	P14.1	P15.4	P15.2	VSS	P20.14	B
C	P50_0	P50_1	VSSEXT	P11.12 REFCLK	P11.3 TXD0	P11.15	P11.13	P12.0	P14.6	P14.4	P14.3	VSS	P20.13	P20.10	C
D	P50_2	P50_3	NC	VSSEXT	VFLEX	P11.11 ETHCRS CRSDV	NC	NC	P14.2	P15.5	VSS	P20.9	P20.12	P20.11	D
E	P50_4	P50_5	P10.1	P10.6	VSSEXT					VSS	ESR1 (N)	P20.7	PORST (N)	ESR0 (N)	E
F	P50_6	P50_7	P10.2	P10.5		VEXT	VDDP3	VEXT	VDD		P20.6	P20.8	P21.6, TDI	P21.7 TDO DAP2	F
G	P50_8	P50_9	P10.8 REQ	P10.3 REQ		VEXT	VSS	VSS	VDD		TMS DAP1	P20.3	P20.2, TESTMODE (N)	TCK DAP0	G
H	P50_10	P50_11	P10.7 REQ	P02.1 REQ		VDD	VSS	VSS	VEVRSB		NC	P21_0	P20.0	P21.5	H
J	P02.4	P02.5	P02.2	NC		VSS	VDD	VDD out VDD	VEXT		NC	NC	P21.4	P21.3	J
K	P02.0	P02.7	NC	NC	VSS					VSSEXT	NC	TRST (N)	P21_2	VSS OSC	K
L	P02.8	P02.6	NC	NC	NC	AN2	NC	P33.4	P33.8 FSP	P33.7	VSSEXT	NC	XTAL2	XTAL1	L
M	P02.3	P00.0	NC	AN12	NC	AN3	NC	P33.1	P33.0	P33.12	P33.13	VSSEXT	VDDOSC	VEXT OSC	M
N	NC	NC	NC	AN11	AN8	AN0	NC	P33.3	P33.5	P33.11	P33.10	VCAP1 VGATEP	VSSEXT	P23.1	N
P	NC	VSSM VAGND1	VDDM	VAREF1	AN10	AN1	NC	P33.2	P33.6	P33.9	VEXT	VCAP0 VGATEN	P32.4	NC	P
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	

Figure 21. TC356 ADAS I/O configurations

See references: [1][2][8][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region. The truth table is illustrated later in this document.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal.
- Exceptional case:
  - Ball B3 from group 4 and ball C2 from group 5 are the only pair (indicated by the box with a red border) from adjacent groups which are not sufficiently separated. Therefore it is recommended not to use them together for the monitor signal and mission signal.

*Note: The additional information for choosing the mission and the monitor signals are documented in the appendix; **Additional hint for TC356, BGA180 package***

2.4.3 TC336 ADAS I/O ball configuration in the LFBGA180 package variant

Note: Regions of different colour indicate the different groups for the mission and monitor signals.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
A	NC	NC	P11.10 RXD0	P11.9 RXD1	P11.6 TXEN	P11.2 TXD1	P11.0 TXD3	P12.1	P14.0	P15.3	P15.6	P15.1	P15.0	NC	A
B	NC	VSSEXT	P11.8 RXD2	P11.7 RXD3	P11.5 TXCLK	P11.4 RXCLK	P11.1 TXD2	P11.14	P14.5	P14.1	P15.4	P15.2	VSS	P20.14	B
C	P50_0	P50_1	VSSEXT	P11.12 REFCLK	P11.3 TXD0	P11.15	P11.13	P12.0	P14.6	P14.4	P14.3	VSS	P20.13	P20.10	C
D	P50_2	P50_3	NC	VSSEXT	VFLEX	P11.11 ETHCRS CRSDV	NC	NC	P14.2	P15.5	VSS	P20.9	P20.12	P20.11	D
E	P50_4	P50_5	P10.1	P10.6	VSSEXT					VSS	ESR1 (N)	P20.7	PORST (N)	ESR0 (N)	E
F	P50_6	P50_7	P10.2	P10.5		VEXT	VDDP3	VEXT	VDD		P20.6	P20.8	P21.6, TDI	P21.7 TDO DAP2	F
G	P50_8	P50_9	P10.8 REQ	P10.3 REQ		VEXT	VSS	VSS	VDD		TMS DAP1	P20.3	P20.2, TESTMODE (N)	TCK DAP0	G
H	P50_10	P50_11	P10.7 REQ	P02.1 REQ		VDD	VSS	VSS	VEVRSB		NC	P21_0	P20.0	P21.5	H
J	P02.4	P02.5	P02.2	AN17		VSS	VDD	VDD out VDD	VEXT		NC	NC	P21.4	P21.3	J
K	P02.0	P02.7	AN20	AN16	VSS					VSSEXT	NC	TRST (N)	P21_2	VSS OSC	K
L	P02.8	P02.6	NC	AN9	AN4	AN2	NC	P33.4	P33.8 FSP	P33.7	VSSEXT	NC	XTAL2	XTAL1	L
M	P02.3	P00.0	AN21	AN12	AN5	AN3	NC	P33.1	P33.0	P33.12	P33.13	VSSEXT	VDDOSC	VEXT OSC	M
N	AN15	AN14	AN13	AN11	AN8	AN0	NC	P33.3	P33.5	P33.11	P33.10	VCAP1 VGATEP	VSSEXT	P23.1	N
P	NC	VSSM VAGND1	VDDM	VAREF1	AN10	AN1	NC	P33.2	P33.6	P33.9	VEXT	VCAP0 VGATEN	P32.4	NC	P
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	

Figure 22. TC336 ADAS I/O configurations

See references: [1][2][10][12]

- Eight different regions are highlighted to indicate that balls (also routing and pads) from one region are sufficiently separated from balls (routing and pads) from another region. The truth table is illustrated later in this document.
- To reduce the risk of CCFs leading to the failure of both mission and monitor signals, it is recommended to use the balls from one region for the mission signal and the ball from another region for the monitor signal.
- <u>Exceptional case</u>:
  - Ball B3 from group 4 and ball C2 from group 5 are the only pair (indicated by the box with a red border) from adjacent groups which are not sufficiently separated. It is therefore recommended not to use them together for the monitor signal and mission signal.

Note**: The additional information for choosing the mission and the monitor signals are documented in the appendix; **Exceptions for TC336 ADAS,** **BGA180 package

2.5 Re-distribution layer limitations

- The layout of each LFBGA package is an individual design.
- The differences in the routing can cause dedicated limitations.
- The recommendation given above are valid for all the listed products of the AURIX™ platform.
  - TC399 LFBGA516 package
  - TC389 LFBGA516 package
  - TC397 LFBGA292 package
  - TC397 LFBGA292 ADAS package
  - TC387 LFBGA292 package
  - TC377 LFBGA292 package
  - TC377 TX LFBGA292 package
  - TC377 TE LFBGA292 package
  - TC367 LFBGA292 package
  - TC357 LFBGA292 package
  - TC337 LFBGA292 package
  - TC337 EXT LFBGA292 package
  - TC3E7 LFBGA292 package
  - TC366 LFBGA180 package
  - TC336 LFBGA180 package
  - TC356 LFBGA180 ADAS package
  - TC336 LFBGA180 ADAS package

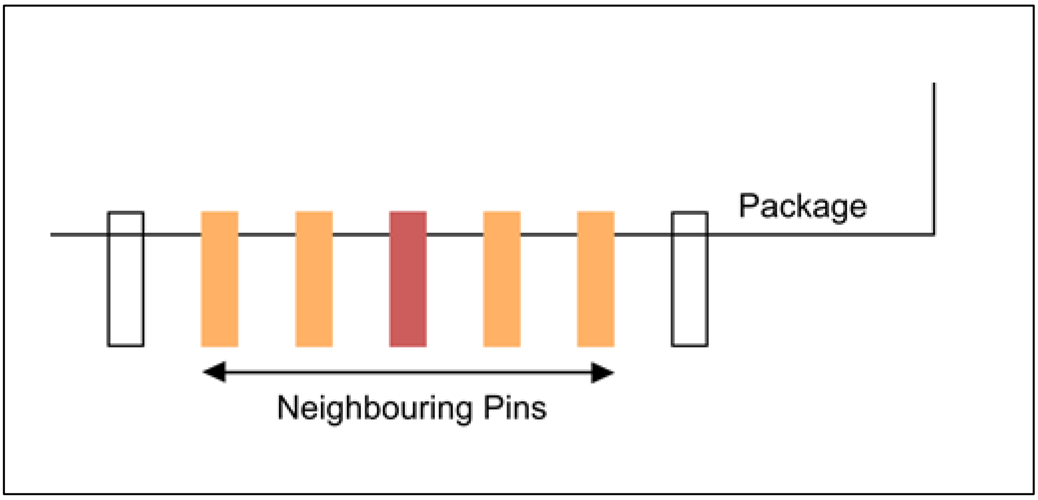
3 Common Cause Failures in QFP package

The CCF can occur at one of the following parts of the package:

1. Neighboring pins

   This illustrative example shows how a common cause failure can affect the neighbouring balls at the package level.

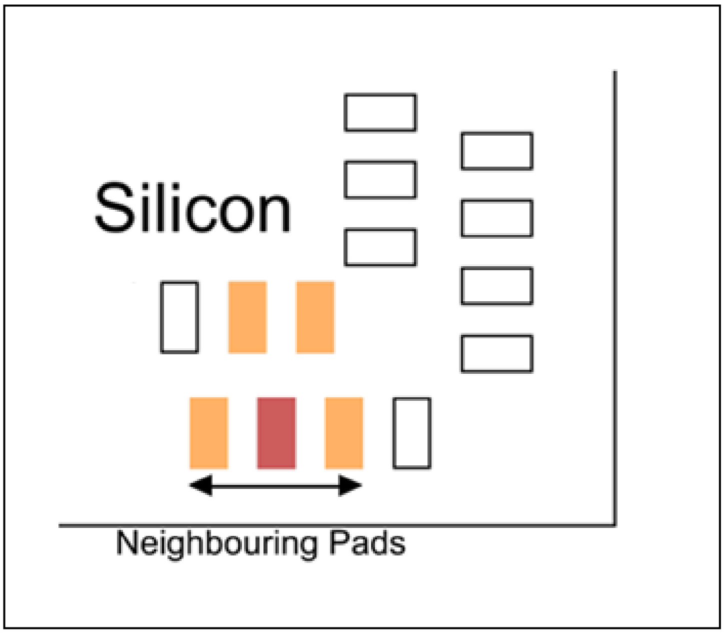
   The pin in consideration is marked in red color while the affected balls are marked in orange color.



2. Neighbouring pads

   This illustrative example how a common cause failure can affect the neighboring pads on the Silicon.

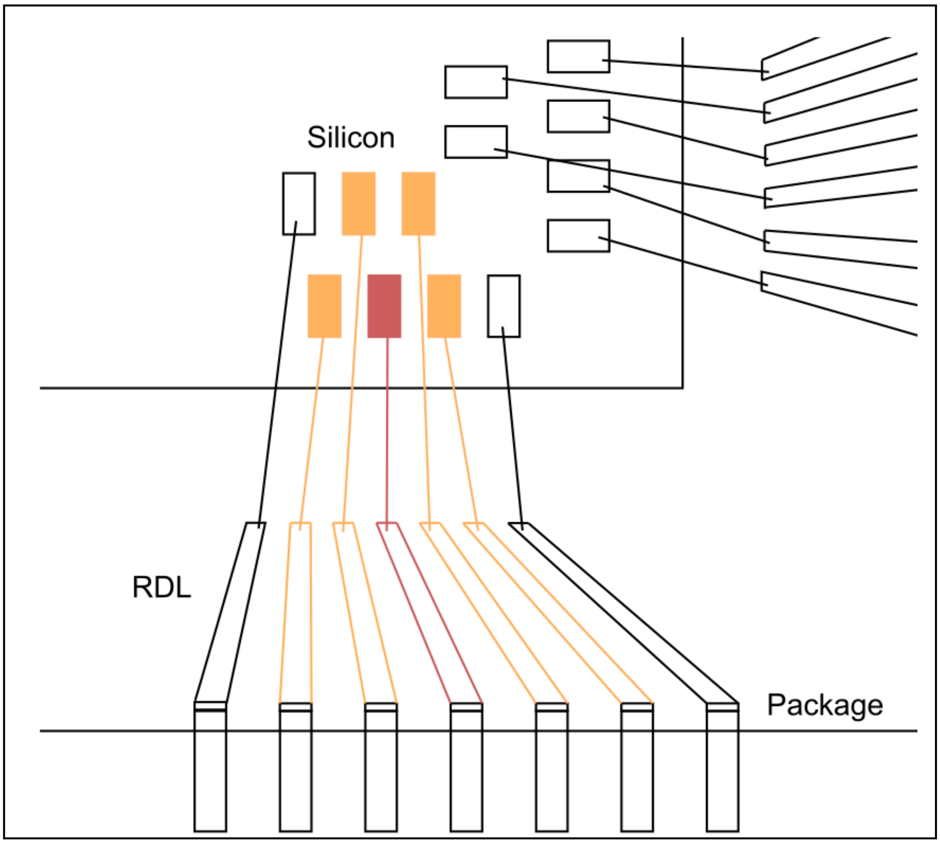
   The pad in consideration is marked in red color while the affected pads are marked in orange color.



3. In the package redistribution layer

This illustrative example shows how a common cause failure can propogate through the re-distribution (RDL) layer and can affect other components as well.

The substrate routing in the re-distribution layer in consideration is marked in red color while the affected RDL routings are marked in orange color.



Note: *It is important to understand that the neighboring pins are not always routed to neighboring pads. Therefore the proximity between the mission and monitor input/output signal at the pins, redistribution layer, and pads, has been considered for the recommendations given.*

3.1 Recommendation for QFP packages

Products with leadframe based packages have different CCF's. The pins are connected to the Die by leadframe in the LQFP package. It is recommended that the mission and monitor signals shall have at least a distance of 3 pins on the package.

This distance between the mission and the monitor signals is to avoid a possible short-circuit on the leadframe or between bond wires in the package. This distance also ensures that shorts between neighboring pins at the package outside do not cause a common cause failure for the mission and the monitor signals.

Example

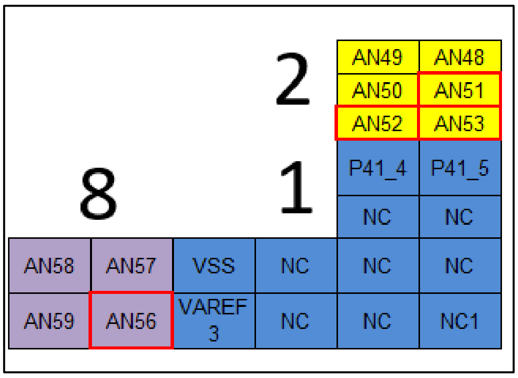
If the mission signal is on the pin number X, the monitor signal should be connected atleast X+3 or X-3 pins away.

4 Appendix

4.1 Exceptions for TC399, BGA516 package

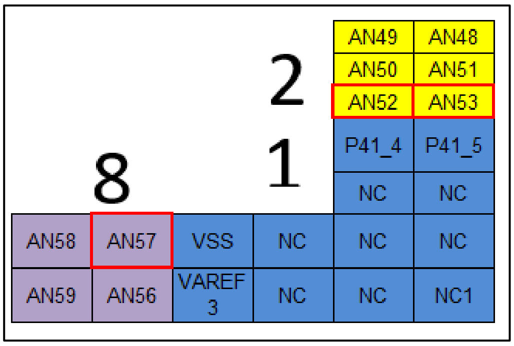
1. AN56 from group 8 should not be used redundantly with AN51, AN52 or AN53 from group 2. In the image, the boxes of these exceptions are indicated with a red border.

   AN56 is not sufficiently separated from the listed AN channels, therefore it is recommended not to use them for the monitor signal and mission signal.



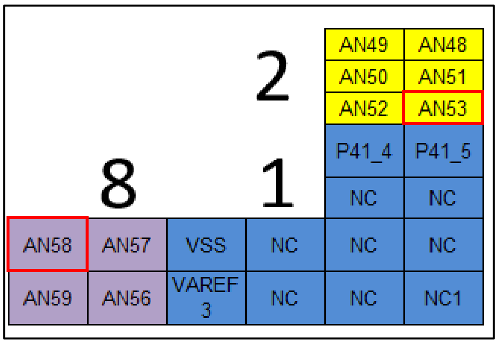
2. AN57 from group 8 should not be used redundantly with AN52 or AN53 from group 2. In the image, the boxes of these exceptions are indicated with a red border.

   AN57 is not sufficiently separated from the listed AN channels, therefore it is recommended not to use them for the monitor signal and mission signal.



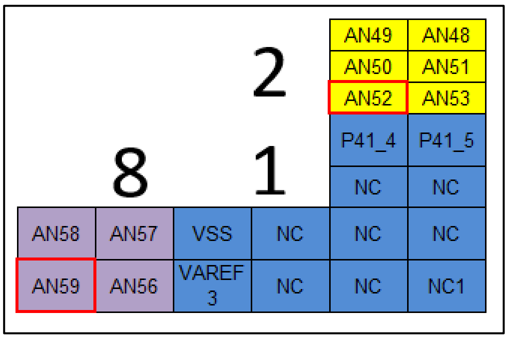
3. AN58 from group 8 should not be used redundantly with AN53 from group 2. In the image, the boxes of these exceptions are indicated with a red border.

   AN58 is not sufficiently separated from the listed AN channels, therefore it is recommended not to use them for the monitor signal and mission signal.



4. AN59 from group 8 should not be used redundantly with AN52 from group 2. In the image, the boxes of these exceptions are indicated with a red border.

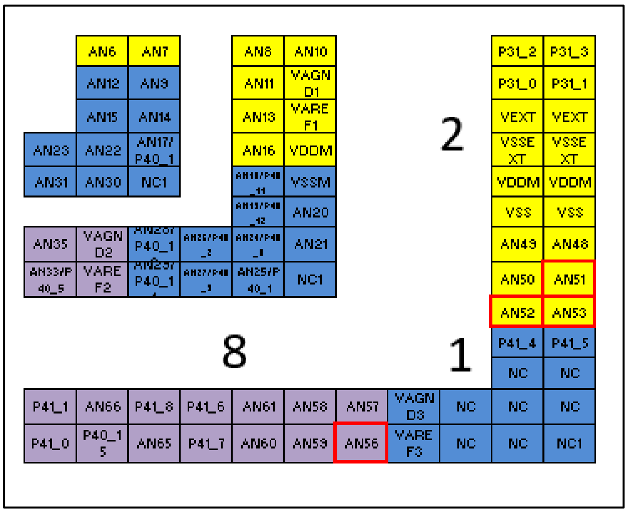
   AN58 is not sufficiently separated from the listed AN channels, therefore it is recommended not to use them for the monitor signal and mission signal.



4.2 Exceptions for TC389, BGA516 package

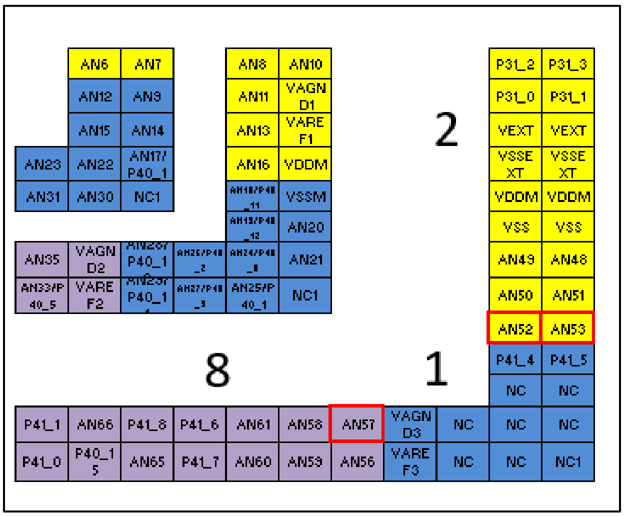
1. AN56 from group 8 should not be used redundantly with AN51, AN52 or AN53 from group 2. In the image, the boxes of these exceptions are indicated with a red border.

   AN56 is not sufficiently separated from the listed AN channels, therefore it is recommended not to use them for the monitor signal and mission signal.



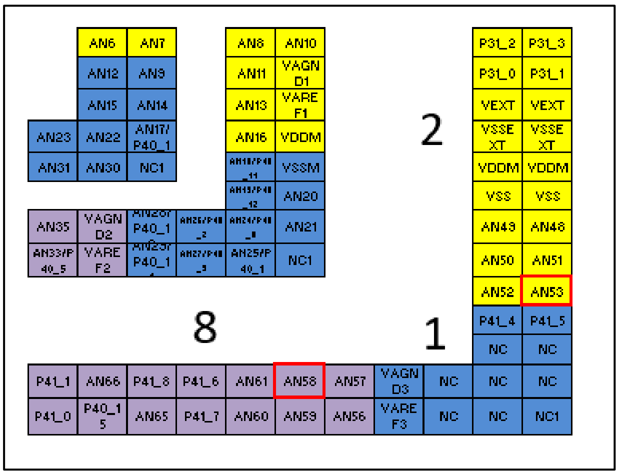
2. AN57 from group 8 should not be used redundantly with AN52 or AN53 from group 2. In the image, the boxes of these exceptions are indicated with a red border.

   AN57 is not sufficiently separated from the listed AN channels, therefore it is recommended not to use them for the monitor signal and mission signal.



3. AN58 from group 8 should not be used redundantly with AN53 from group 2. In the image, the boxes of these exceptions are indicated with a red border.

   AN58 is not sufficiently separated from the listed AN channels, therefore it is recommended not to use them for the monitor signal and mission signal.



4.3 Additional hint for TC357, BGA292 package

This section elaborates the sufficiently separated AN channels.

Note: The redudant AN channel marked as 'TRUE' with respect to the mission AN channel, can be used redundantly.

Monitor AN Channel \ Mission AN Channel	AN0	AN1	AN2	AN3	AN4	AN5	AN6	AN7	AN8	AN9	AN10	AN11	AN12	AN13	AN14	AN15
AN0	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN2	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE
AN3	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN4	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE
AN5	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE
AN6	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE	TRUE	FALSE
AN7	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE	FALSE	FALSE
AN8	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE
AN9	TRUE	TRUE	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE
AN10	TRUE	TRUE	FALSE	TRUE	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE
AN11	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE
AN12	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
AN13	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE
AN14	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
AN15	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE

Figure 21. Truth table with valid AN channels for mission and monitor AN channels

4.4 Additional hint for TC337 EXT, BGA292 package

This section elaborates the sufficiently separated AN channels.

Note: The redudant AN channel marked as 'TRUE' with respect to the mission AN channel, can be used redundantly.

Monitor AN Channel	Mission AN Channel																							
	AN0	AN1	AN2	AN3	AN4	AN5	AN6	AN7	AN8	AN9	AN10	AN11	AN12	AN13	AN14	AN15	AN16	AN17	AN18	AN19	AN20	AN21	AN22	AN23
AN0	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN2	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN3	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN4	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN5	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN6	FALSE	TRUE	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN7	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN8	TRUE	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN9	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN10	TRUE	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN11	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN12	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN13	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
AN14	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
AN15	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE
AN16	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE
AN17	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE
AN18	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
AN19	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
AN20	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
AN21	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
AN22	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
AN23	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE

Figure 22. Truth table with valid AN channels for mission and monitor AN channels

4.5 Additional hint for TC356, BGA180 package

This section elaborates the sufficiently separated AN channels.

Note: The redudant AN channel marked as 'TRUE' with respect to the mission AN channel, can be used redundantly.

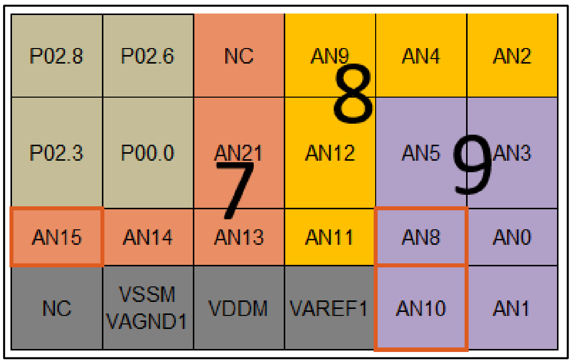
Monitor AN Channel	Mission AN Channel							
	AN0	AN1	AN2	AN3	AN8	AN10	AN11	AN12
AN0	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE
AN1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE
AN2	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
AN3	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE
AN8	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
AN10	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	TRUE
AN11	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
AN12	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE

Figure 23. Truth table with valid AN channels for mission and monitor AN channels

4.6 Exceptions for TC336 ADAS, BGA180 package

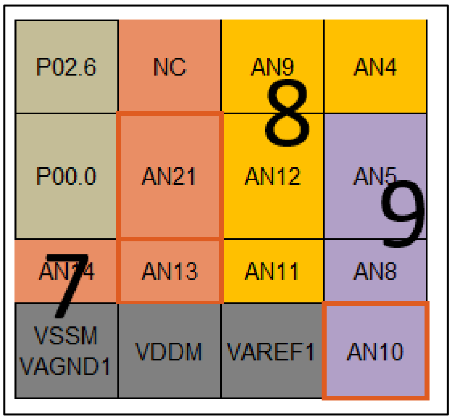
1. AN15 from group 7 should not be used redundantly with AN8, or AN10 from group 9. In the image, the boxes of these exceptions are indicated with a red border.

   AN15 is not sufficiently separated from the listed AN channels, therefore it is recommended not to use them for the monitor signal and mission signal.



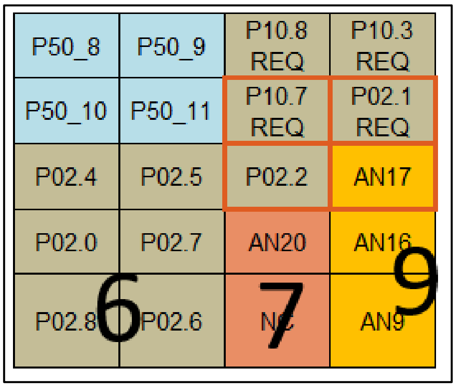
2. AN10 from group 9 should not be used redundantly with AN13, or AN21 from group 7. In the image, the boxes of these exceptions are indicated with a red border.

   AN10 is not sufficiently separated from the listed AN channels, therefore it is recommended not to use them for the monitor signal and mission signal.



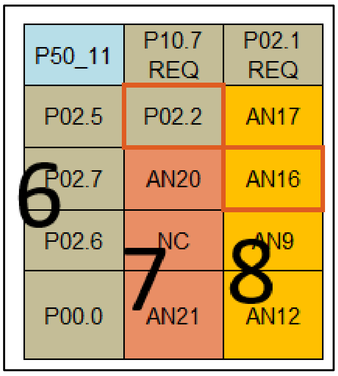
3. AN17 from group 9 should not be used along with P10.7, P2.1 and P2.2 from group 6. In the image, the boxes of these exceptions are indicated with a red border.

   AN17 is not sufficiently seperated from the listed pins.



4. AN16 from group 9 should not be used along with P2.2 from group 6. In the image, the boxes of these exceptions are indicated with a red border.

   AN16 is not sufficiently seperated from the listed pins.



5 References

[1] Infineon Technologies AG, AURIX™ Safety Manual V1.12

[2] International Standard ISO26262, ISO 26262-1:2011

[3] Infineon Technologies AG, Datasheet TC39x 32-bit single chip microcontroller, Munich: Infineon Technologies AG.

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[10] Infineon Technologies AG, Datasheet TC33xEXT 32-bit single chip microcontroller, Munich: Infineon Technologies AG.

[11] Infineon Technologies AG, Datasheet TC3Ex 32-bit single chip microcontroller, Munich: Infineon Technologies AG.

[12] IEC CEI 60664-5 Insulation co-ordination for equipment within low-voltage systems, International Electrotechnical commission, 2007-04.

6 Revision history

Major changes since the last revision

Version	Date	Description of change
1.0	2018-03	Initial Version
2.0	2019-07	- BGA516 packages added for TC39x and TC38x. - BGA292 packages added for TC37x, TC36x, TC35x and TC33x - BGA180 packages added for TC36x, TC35x and TC33x. - Recommendations added for Quad Flat Packages
2.1	2019-08	- Typo errors fixed in explanation for depiction of CCF for BGA (Page 4) Neighboring pads and CCF for QFP packages (page 28) - Amended the Truth Table & Grouping for BGA180 ADAS - Exceptions added for TC336, BGA180 package
2.2	2021-02	- BGA292 packages added for TC337EXT and TC3E7 - Typo errors fixed

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